CENTRAL MANUFACTURING TECHNOLOGY INSTITUTE

C – Smart Manufacturing Cell

User Guide

Troubleshooting-

MDB Data Management

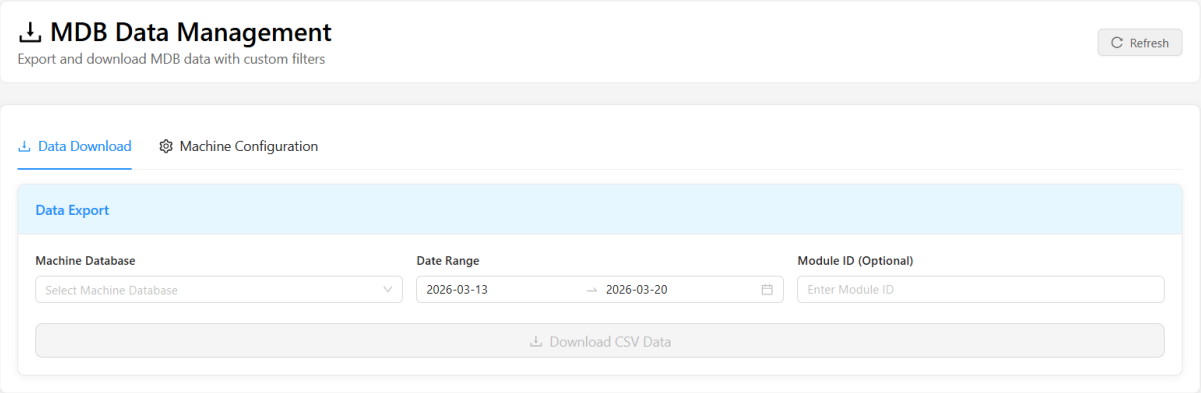
PROJECT: SPV Data Flow

MDB Data Management Issues

The following section addresses the most common issues encountered while using the MDB Data Management module – including data download failures and machine connection errors.

Download Button Disabled

If the Download CSV Data button cannot be clicked:

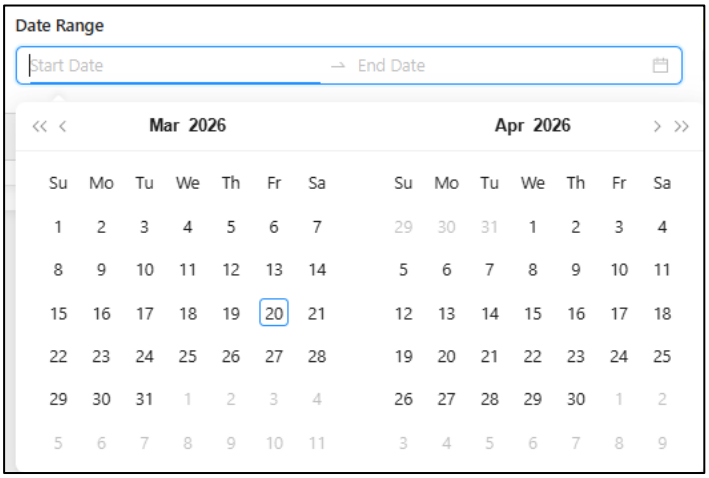


Step 1: Select a Machine Database

→ Click the Machine Database dropdown and choose a valid MDB machine. The button remains disabled until a machine is selected.

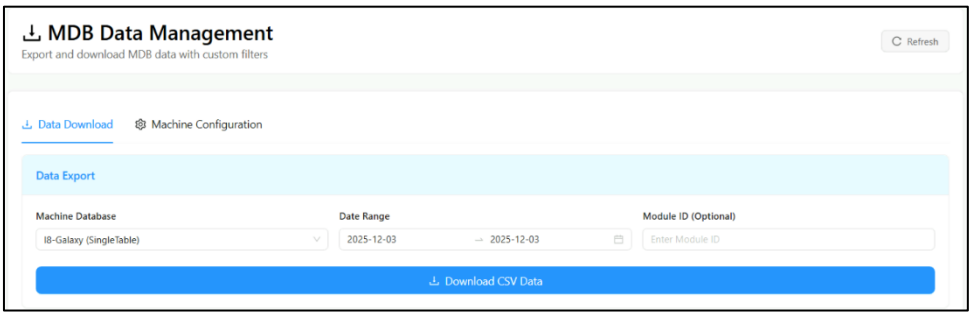
Step 2: Set the Date Range

→ Ensure both start and end dates are filled in the Date Range picker. The button activates only when both fields have valid dates.



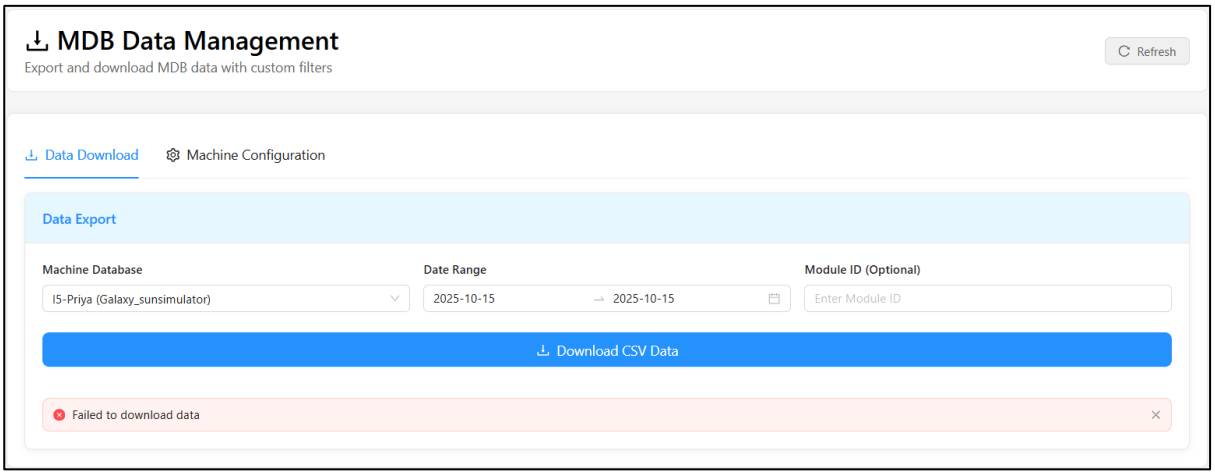
Step 3: **Attempt Download**

→ Once the machine and date range are selected, the Download CSV Data button turns blue. Click it to begin the export.



Failed to download data Error

If a red error banner appears after clicking the download button:



Step 1: **Verify data exists for the selected date range**

→ Confirm with the machine operator that data was recorded during the chosen period. Try a wider date range if unsure.

Step 2: **Test the machine connection**

→ Navigate to the Machine Configuration tab and run a connection test on the selected machine.

Machine Configuration – Connection Test

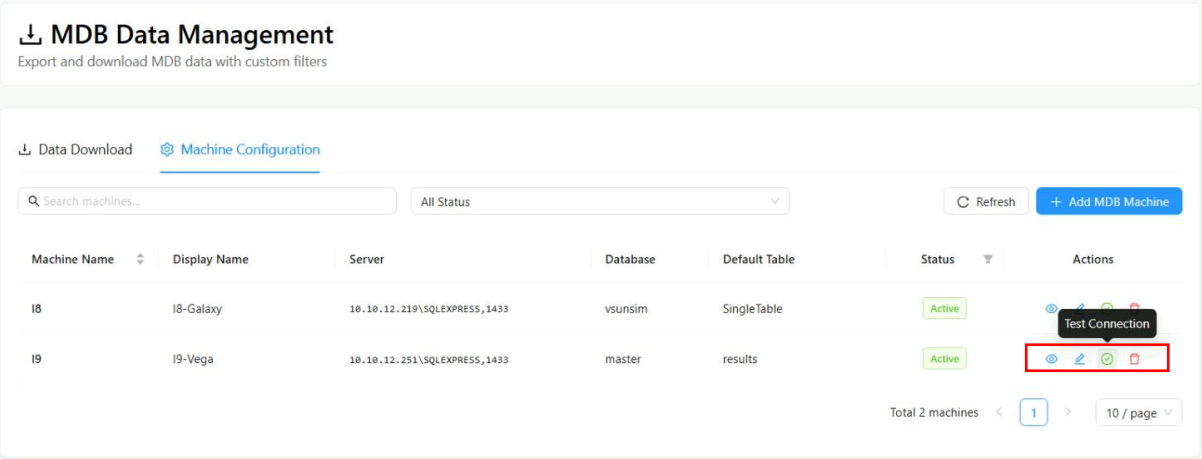
To verify machine connectivity, navigate to the Machine Configuration tab.

Step 1: **Open Machine Configuration**

→ Click the Machine Configuration tab. A list of all registered MDB machines is displayed with their server details and status.

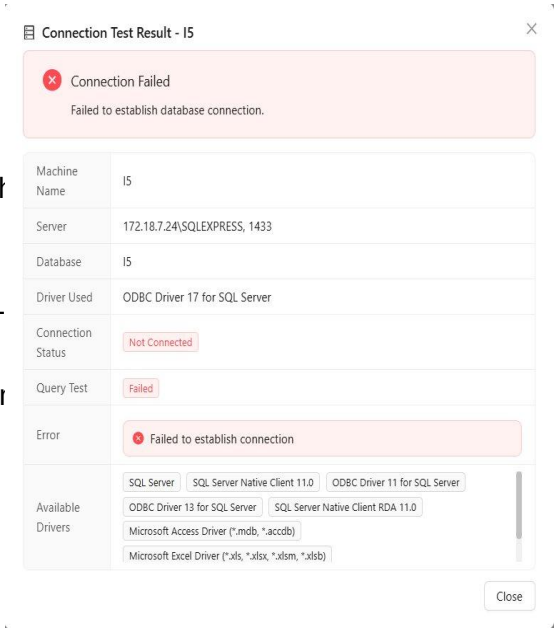
Step 2: Run Test Connection

→ Click the Test Connection icon (✓) in the Actions column for the relevant machine. A result dialog will appear.



Connection Test Result – "Connection Failed"

If the dialog shows Connection Failed – Failed to establish database connection:



Step 1: Note the error details

Record the Server IP, Port, Database name, and tl error message shown in the dialog.

Step 2: Verify configuration

Ensure the Server field follows the format: IP\INST ANCENAME, PORT (e.g. 10.10.12.219\SQL EXPRESS, 1433). If incorrect, click the Edit icon ar update it.

Step 3: Check ODBC Driver

The application requires ODBC Driver 17 for SQL Server. If it is missing, install it from the Microsoft Download Center.

Network Reachability – Ping Test

If the connection test fails, check whether the machine is reachable on the network.

Step 1: Open Command Prompt as Administrator

→ Press Win + R, type cmd, and run as Administrator.

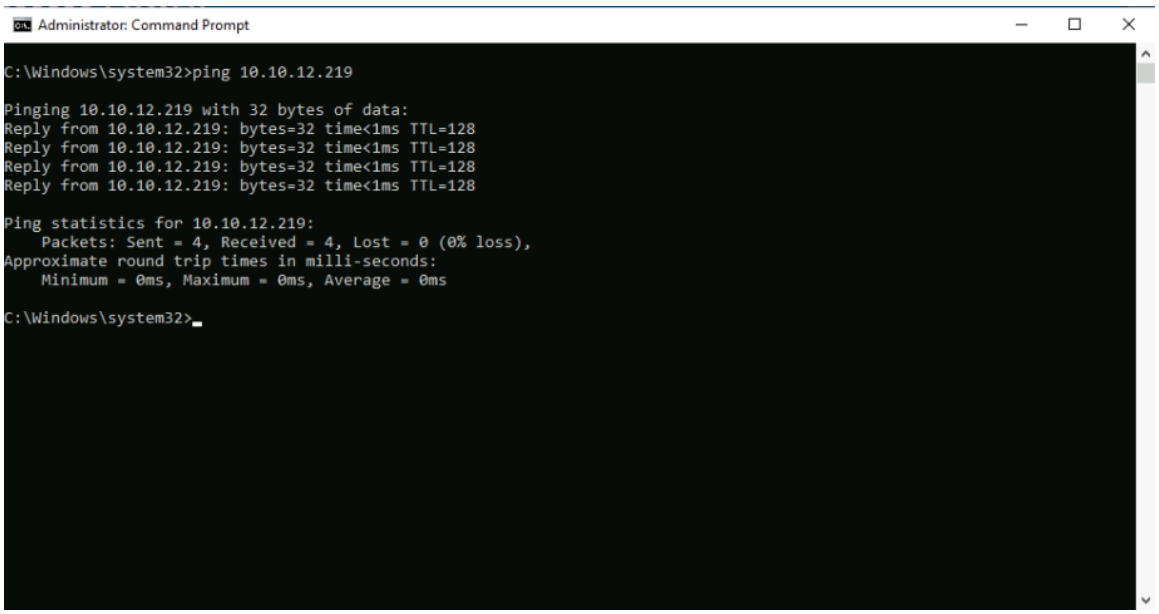
Step 2: Ping the machine IP

→ Run the following command, replacing the IP with your machine's address:

```
ping < machine_ip>
```

Step 3: Interpret the result

Ping Result	Meaning	Action
Reply from IP: bytes=32 TTL=128	Host reachable. Network OK.	Proceed to SQL Server checks (3.6).
Request timed out	Firewall blocking or machine off.	Check firewall / verify machine is on.
Destination unreachable	No network route to host.	Contact network admin for routing fix.



→ The screenshot above confirms 10.10.12.219 is reachable (0% packet loss, TTL=128).

SQL Server Troubleshooting (Ping Passes, Connection Still Fails)

Step 1: Verify SQL Server service is running

→ On the target machine, open Services (services.msc). Locate SQL Server (SQLEXPRESS) and ensure its status is Running. Also confirm SQL Server Browser is Running – this is required for named instances.

Step 2: Enable TCP/IP in SQL Server Configuration Manager

→ Open SQL Server Configuration Manager → SQL Server Network Configuration → Protocols for SQLEXPRESS. Enable TCP/IP and set the TCP Port to 1433 under IPAll. Restart the SQL Server service after changes.

Step 3: Allow port 1433 through Windows Firewall

→ Open Windows Firewall with Advanced Security → Inbound Rules. Create a new rule: Port → TCP → 1433 → Allow the connection → apply to all profiles.

Step 4: Verify the database name

→ Open SQL Server Management Studio (SSMS) and confirm the database (e.g. I8) exists. If the name is wrong in the application, edit the machine in Machine Configuration and correct it.

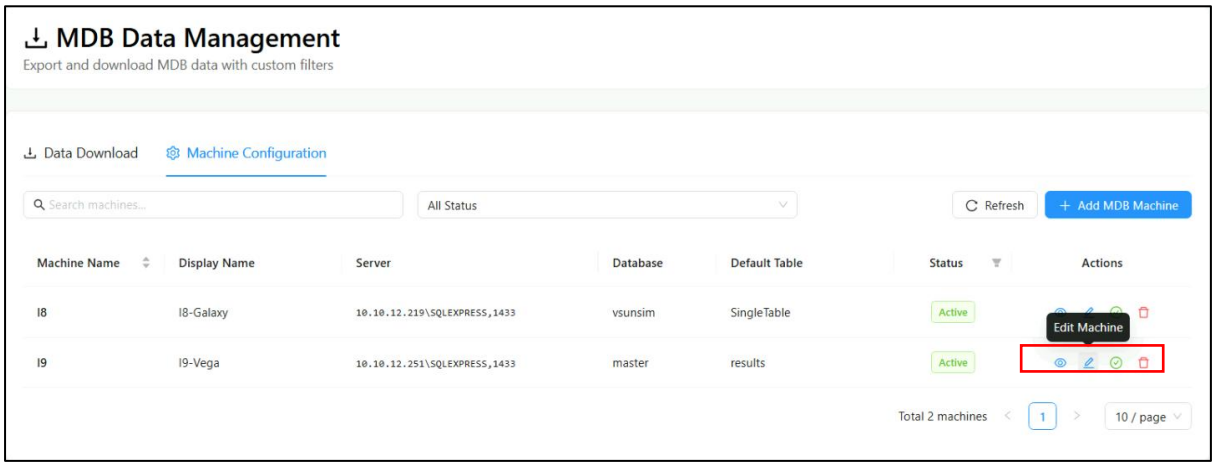
Step 5: Verify the expiry of database credentials

→ Open SQL Server Management Studio (SSMS) and confirm if the login credentials have expired or not. If expired, reset password

Editing Machine Configuration

To correct connection details for an existing machine:

- Navigate to the Machine Configuration tab.
- Click the Edit (pencil) icon in the Actions column for the relevant machine.



- Update incorrect fields: Server IP, Port, or Database Name.
- Save changes, then re-run Test Connection to confirm.